

ELROY GOH

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PROFILE

Quantitative researcher with 5+ years' experience developing and deploying systematic trading strategies in energy and commodity markets. Specialised in alpha signal design, regime-aware validation, and execution-aware implementation. Experience spans the full research lifecycle from signal discovery and backtesting to production deployment and live trading.

EXPERIENCE

Desk Quant / Trading Assistant · Gunvor Group Ltd

Singapore | Jun 2021 – Sep 2025

Alpha Research & Strategy Development

- Developed and deployed systematic trading strategies across energy markets, combining mean-reversion, trend-following, and fundamental signals
- Evaluated signals using information coefficient (IC), turnover, and capacity constraints, with regime-conditioned performance analysis prior to live deployment
- Analysed signal decay, robustness, and cross-market generalisation using walk-forward and out-of-sample validation frameworks
- Researched academic and industry literature (e.g. carry, momentum, basis mean-reversion) to source and refine new alpha signals
- Executed OTC derivatives trades (100–500 kb/d exposure) across crude and refined products, translating systematic signals into trade execution under liquidity, timing, and market impact constraints

Trading Infrastructure & Engineering

- Built end-to-end automation pipeline for real-time signal ingestion, order routing, position reconciliation, and exception handling across multiple exchanges
- Optimised proprietary VNPY-based execution platform, resolving race conditions and eliminating order/trade state inconsistencies during high-frequency updates
- Implemented real-time monitoring and alerting systems for fills and strategy halts, improving execution reliability and reducing unmonitored outages
- Developed execution algorithms for energy futures trading across fragmented liquidity venues
- Designed and maintained data infrastructure (PostgreSQL, MongoDB) for tick-level and daily market data, enabling efficient research and backtesting workflows
- Mentored junior trading assistants on execution systems and operational workflows

Risk Analyst · Olam International

Singapore | Nov 2019 – Jun 2021

- Researched and implemented risk models (VaR, CVaR, Greeks) for commodity trading portfolios, with backtesting and model validation
- Built Python-based dashboard for real-time risk monitoring (P&L attribution, delta, VaR) across trading books
- Supported migration to new database infrastructure for risk systems, coordinating across trading, development, and data teams

EDUCATION

MSc Quantitative Finance

Aug 2025 – Jul 2026

Singapore Management University · cGPA 4.0 / 4.0

Modules: Stochastic Modelling, Derivatives, Quantitative Trading Strategies, Econometrics, Portfolio Management, Market Microstructure & Algorithmic Trading

MSc Investment and Wealth Management

Aug 2016 – Jul 2017

Imperial College Business School, London · Distinction in core modules

Modules: Asset Allocation & Investment Strategies, Derivatives, Financial Econometrics, Portfolio Management, Mathematics for Finance

BSc Accounting & Finance, First Class Honours

Sep 2013 – Jun 2016

University of Southampton · Deloitte Management Prize (top 2nd-year undergraduate)

SKILLS

Quantitative Methods Signal research & alpha generation; factor construction & decay analysis; time-series modelling (AR, GARCH, cointegration, regime detection); statistical arbitrage; backtesting & walk-forward validation; VaR / CVaR; copula methods; Monte Carlo simulation; options pricing (BSM, SABR, Displaced Diffusion)

Programming Python (NumPy, pandas, SciPy, statsmodels, scikit-learn, PyTorch); VNPY; SQL; PostgreSQL; MongoDB

Data & Infrastructure Tick and daily market data pipelines; real-time signal ingestion; automated order routing; Bloomberg Terminal; Refinitiv Datastream

CERTIFICATIONS & ACHIEVEMENTS

Financial Risk Manager (FRM) · Passed Levels I & II · GARP

CFA Level I · Passed

1st Place — JP Morgan Business Challenge 2015 · 1st Prize — PwC Financial Challenge 2015

Languages: English (native), Mandarin (native), Cantonese (conversational) · Work authorisation: Singapore Citizen